

Write R code that lasts

Scientific workflows: Tools and Tips 

2025-05-15

What is this lecture series?

Scientific workflows: Tools and Tips

 Every 3rd Thursday  4-5 p.m.  Webex

- One topic from the world of scientific workflows
- Material provided [online](#)
- If you don't want to miss a lecture
 - [Subscribe to the mailing list](#)

The problem



Artwork by [Allison Horst](#), CC BY 4.0

Selina Baldauf // Reproducible data analysis

The problem with messy projects

 **Maintainability** Can I understand, update and fix my own code?

 **Reproducibility** Can someone else reproduce my results?

 **Reliability** Will my code work in the future?

 **Reusability** Can someone else actually use my code?

Today

Learn how to keep the kitchen clean.

- Basic tips for **reproducible** and **maintainable** data analysis projects
 - Project organization
 - Coding
 - Dependency management

The reproducibility mindset

- Imagine how someone else will see the project for the first time.

Will they be able to understand and use your project?

Practical reproducibility check

Send your project to a colleague and ask them **understand** and **run** the code your analysis.

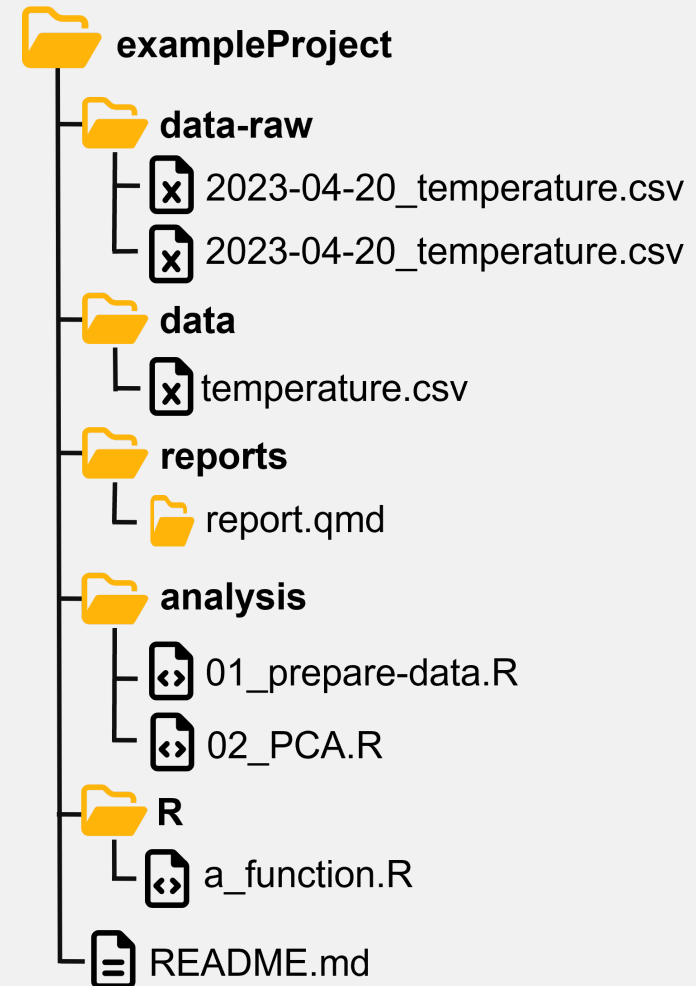
First things first:



Organizing projects

Have a solid project structure

- **Self-contained** projects: code, data, reports, etc. in one place
- Separate things into folders
- Always include a **README** file
- Use standard templates (see e.g. the **template R package**)



Name your files properly

Your collaborators and your future self will love you for this.

Principles ¹

File names should be

1. Machine readable
2. Human readable
3. Working with default file ordering

1. Machine readable file names

Names should allow for easy **searching**, **grouping** and **extracting information** from file names.

- No space & special characters
- Use consistent separators

Bad examples ❌

 2023-04-20 temperature göttingen.csv

 2023-04-20 rainfall göttingen.csv

Good examples ✔️

 2023-04-20_temperature-goettingen.csv

 2023-04-20_rainfall-goettingen.csv

2. Human readable file names

Names should be **informative** and reveal the **file content**.

- Use separators to make it readable

Bad examples ❌

 01preparedataforanalysis.R

 01firstscript.R

Good examples ✔

 01_prepare-data-for-analysis.R

 01_lm-temperature-trend.R

3. Default ordering

If you order your files by name, the ordering should make sense:

- (Almost) always put something numeric first
 - Left-padded numbers (01, 02, ...)
 - Dates in YYYY-MM-DD format

Chronological order

 2023-04-20_temperature-goettingen.csv

 2023-04-21_temperature-goettingen.csv

Logical order

 01_prepare-data.R

 02_lm-temperature-trend.R



Let's start coding

Use save paths

To read and write files, you need to tell R where to find them.

Common workflow:

Set **working directory** with `setwd()`, then read files from there:

```
# set the project root as working directory
setwd("C:/Users/Selina/path/that/only/I/have")

# read my data using a path relative to the working directory
temperature <- read_csv("data/temperature.csv")
```

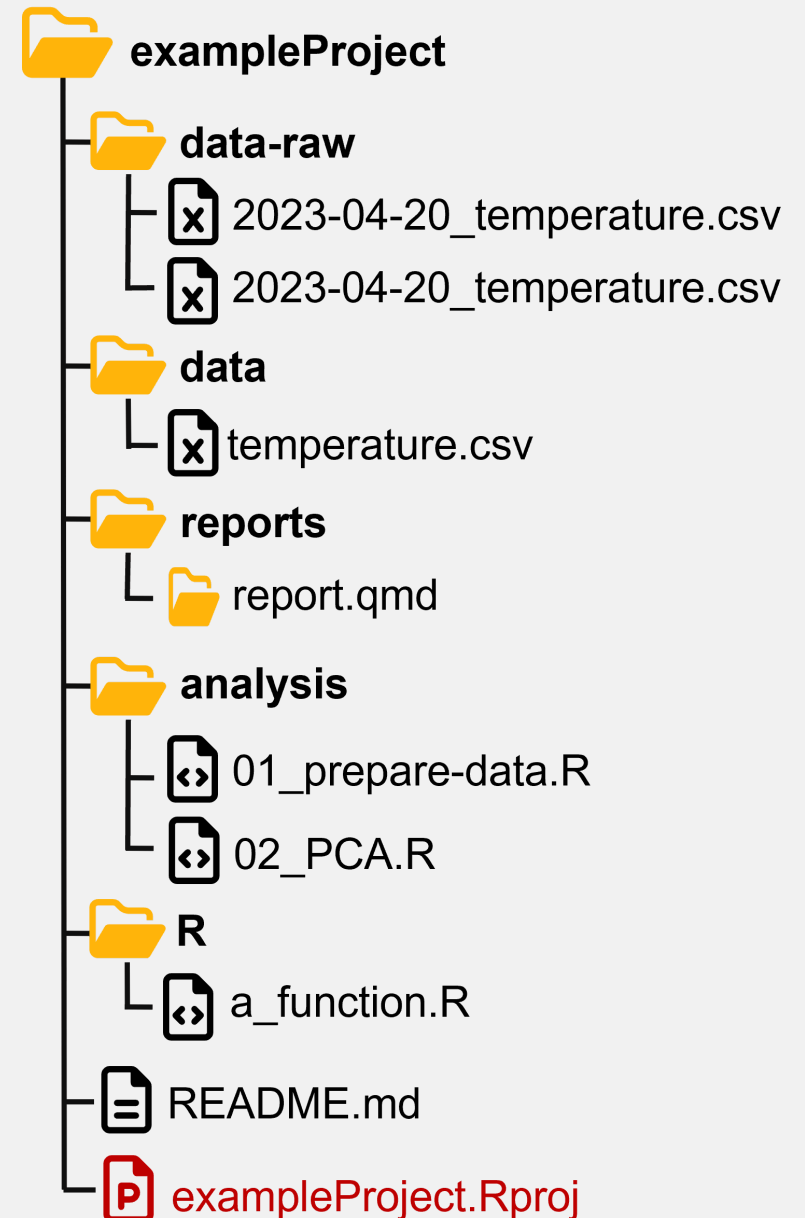
This is **not reproducible!** Your computer at exactly this time is the only one that has this working directory.

Use save paths

Option 1: Use RStudio projects

- Project root is automatically the working directory
- No need to use `setwd()` manually
- Read and write files with relative paths:

```
read_csv("data/temperature.csv")
```



Create an RStudio Project

From scratch:

1. File -> New Project -> New Directory -> New Project
2. Enter a directory name (this will be the name of your project)
3. Choose the directory where the project should be initiated
4. Create Project

Associate an existing folder with an RStudio Project:

1. File -> New Project -> Existing Directory
2. Choose your project folder
3. Create Project

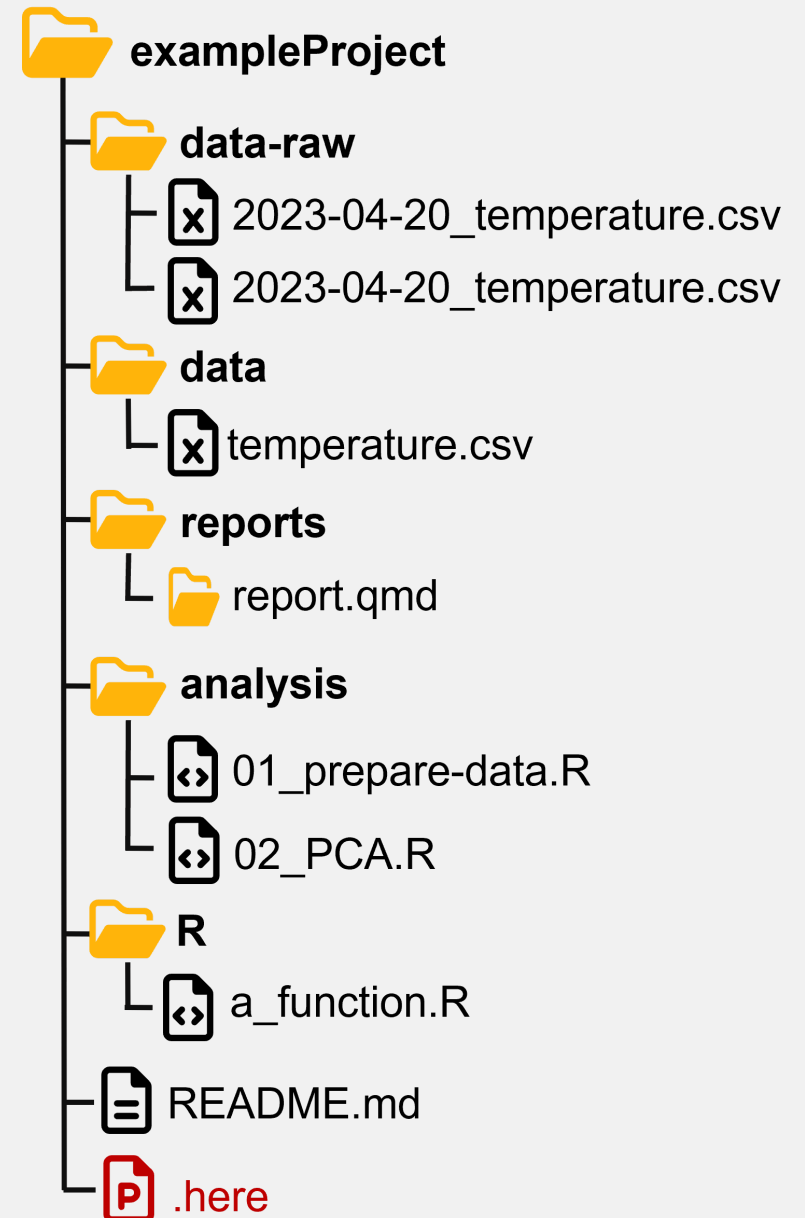
Use save paths

Option 2: Use the **here** package

- Build your paths using the **here** function

```
# use here::here
library(here)
read_csv(here("data", "temperature.csv"))
# or
read_csv(here("data/temperature.csv"))
```

- Reasonable heuristics to find project files
- Looks for ***.Rproj**, **.here**, **.git**, ...
- My recommendation: **always** use **here**
 - with ***.Rproj** OR with **.here**
 - works in all cases



Write well-structured and consistent code



Artwork by [Allison Horst](#), CC BY 4.0

How to structure your scripts

- Use a standardized header
- Initialize at the top
 - `library()` calls
 - global options
 - source additional code
- Read all external data in one place
- Why?
 - fail fast
 - change code in one place

```
# Purpose: Create Figure 2 showing the relationship
# between body mass and bill length
# Authors: Selina Baldauf, Jane Doe, Jon Doe

# load libraries -----
library(tidyverse)
library(vegan)

# Set global options -----
# Plot themes and colors
theme_set(theme_minimal())
custom_colors <- c("cyan4", "darkorange", "purple")

# Source additional code -----
source("R/my_cool_function.R")

# Read data -----
temperature <- read_csv("data/temperature.csv")
rainfall <- read_csv("data/rainfall.csv")
```

How to structure your scripts

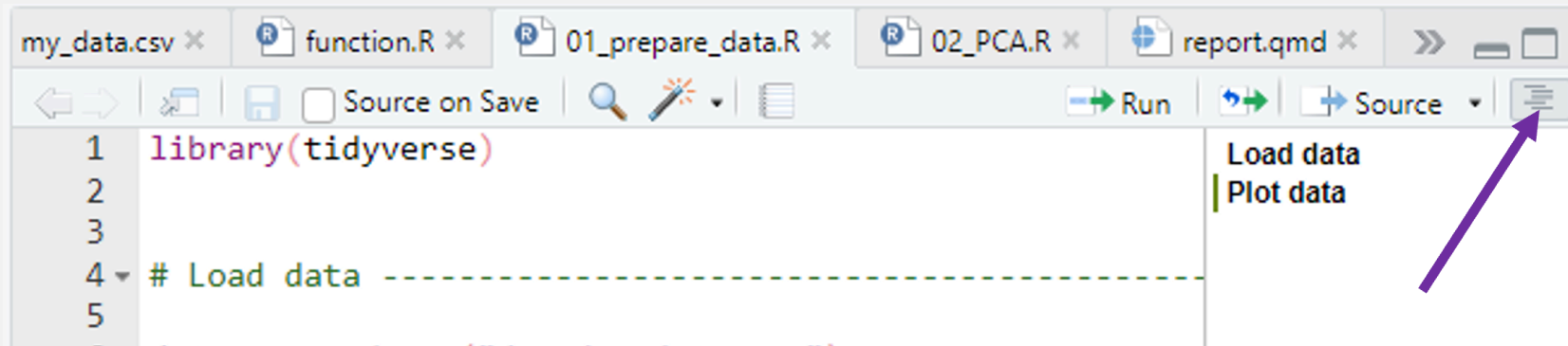
- Use headers to break up your file into sections

```
# Load data -----  
  
input_data <- read_csv(input_file)  
  
# Plot data -----  
  
ggplot(input_data, aes(x = x, y = y)) +  
  geom_point()
```

- Insert a section label with comment + at least 4 ##### or ----
 - RStudio keyboard shortcut: Ctrl/Cmd + Shift + R

How to structure your scripts

- Navigate sections in the file outline



Use a consistent coding style - naming

Have a **naming convention** for variables and functions and stick to it

- Concise and descriptive (nouns for variables, verbs for functions)
- Consistent capitalization: Use **snake_case** for longer variable names
- Avoid conflicts with existing R functions and packages

```
# Good  
day_one
```

```
clean_data()
```

```
# Bad  
dayone  
first_day_of_the_month  
dm1
```

```
f1()
```


Use a consistent coding style - spacing

- Always put spaces after a comma

```
# Good
```

```
x[, 1]
```

```
# Bad
```

```
x[,1]
```

```
x[ , 1]
```

Use a consistent coding style - spacing

- Always put spaces after a comma
- No spaces around parentheses for normal function calls

```
# Good
mean(x, na.rm = TRUE)

# Bad
mean( x, na.rm = TRUE )
mean (x, na.rm = TRUE)
```

Use a consistent coding style - spacing

- Always put spaces after a comma
- No spaces around parentheses for normal function calls
- Spaces around most operators (<-, ==, +, etc.)

```
# Good
height <- (feet * 12) + inches
mean(x, na.rm = TRUE)
```

```
# Bad
height<-feet*12+inches
mean(x, na.rm=TRUE)
```

Use a consistent coding style - spacing

- Always put spaces after a comma
- No spaces around parentheses for normal function calls
- Spaces around most operators (<-, ==, +, etc.)
- Spaces before pipes (|>, %>%) and + in ggplot followed by new line

Good

```
iris|>group_by(Species)|>summarize_if(is.numeric, mean)|>ungroup()
```

Bad

```
iris |>  
  group_by(Species) |>  
  summarize_all(mean) |>  
  ungroup()
```

Use a consistent coding style - spacing

- Always put spaces after a comma
- No spaces around parentheses for normal function calls
- Spaces around most operators (`<-`, `==`, `+`, etc.)
- Spaces before pipes (`|>`, `%>%`) and `+` in ggplot followed by new line

Good

```
ggplot(aes(x = Sepal.Width, y = Sepal.Length, color = Species)) + geom_point()
```

Bad

```
ggplot(aes(x = Sepal.Width, y = Sepal.Length, color = Species)) +  
  geom_point()
```

Use a consistent coding style - width

- Always put spaces after a comma
- No spaces around parentheses for normal function calls
- Spaces around most operators (`<-`, `==`, `+`, etc.)
- Spaces before pipes (`|>`, `%>%`) and `+` in ggplot followed by new line
- Limit your line width to 80 characters.

```
# Bad
iris |> group_by(Species) |> summarise(Sepal.Length = mean(Sepal.Length), Sepal.Width = mean(Sepal.Width), S

# Good
iris |>
  group_by(Species) |>
  summarise(
    Sepal.Length = mean(Sepal.Length),
    Sepal.Width = mean(Sepal.Width),
    Species = n_distinct(Species)
  )
```

Use a consistent coding style

Do I really have to remember all of this?

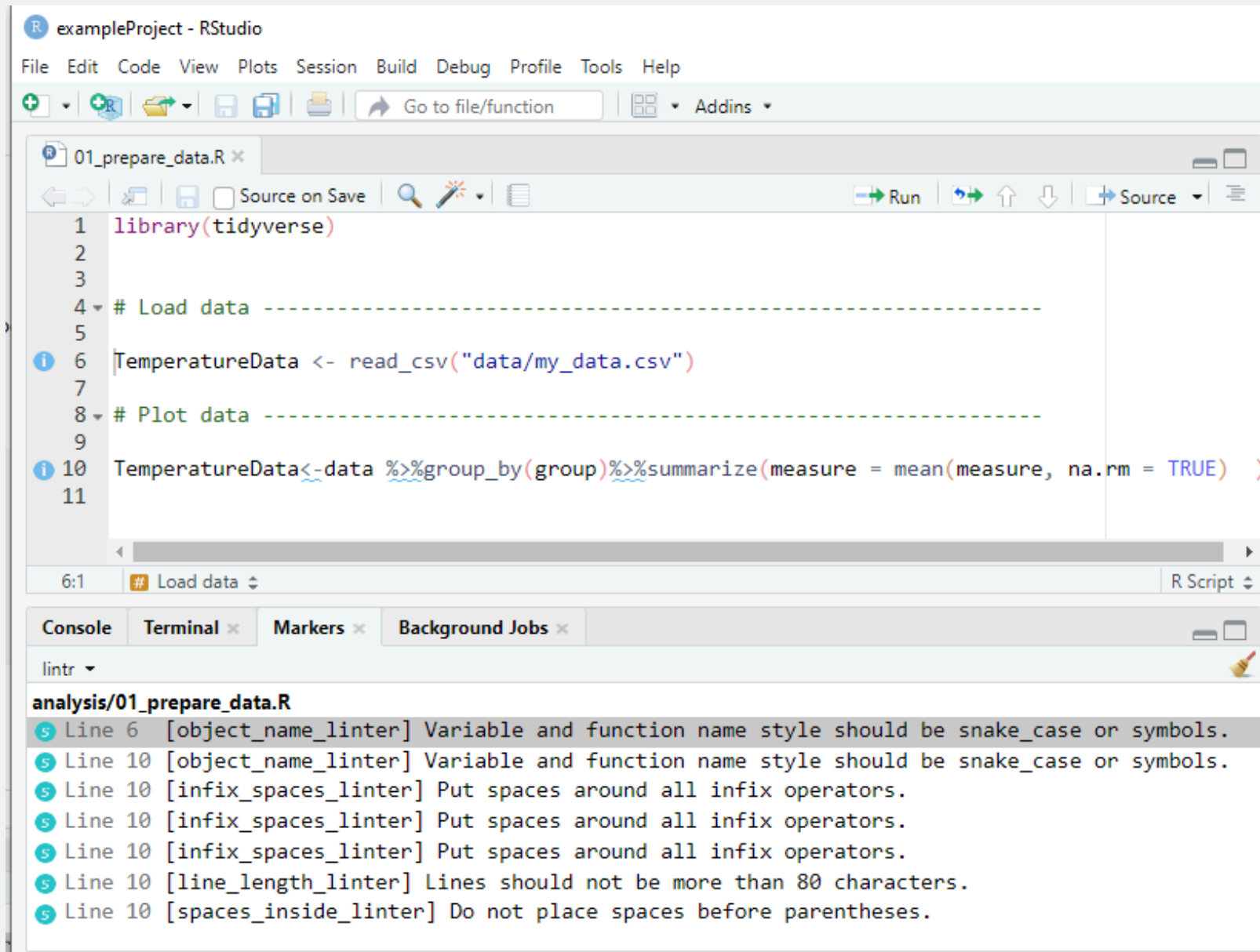
Luckily, no! R and RStudio provide some nice helpers

Coding style helpers - {lintr}

The `lintr` package analyses your code files or entire project and tells you what to fix.

```
# install the package before you can use it
install.packages("lintr")
# lint specific file
lintr::lint(filename = "analysis/01_prepare_data.R")
# lint a directory (by default the whole project)
lintr::lint_dir()
```


Coding style helpers - lintr



The screenshot shows the RStudio interface for a project named 'exampleProject'. The script editor displays a file named '01_prepare_data.R' with the following R code:

```
1 library(tidyverse)
2
3
4 # Load data -----
5
6 TemperatureData <- read_csv("data/my_data.csv")
7
8 # Plot data -----
9
10 TemperatureData <- data %>% group_by(group) %>% summarize(measure = mean(measure, na.rm = TRUE))
11
```

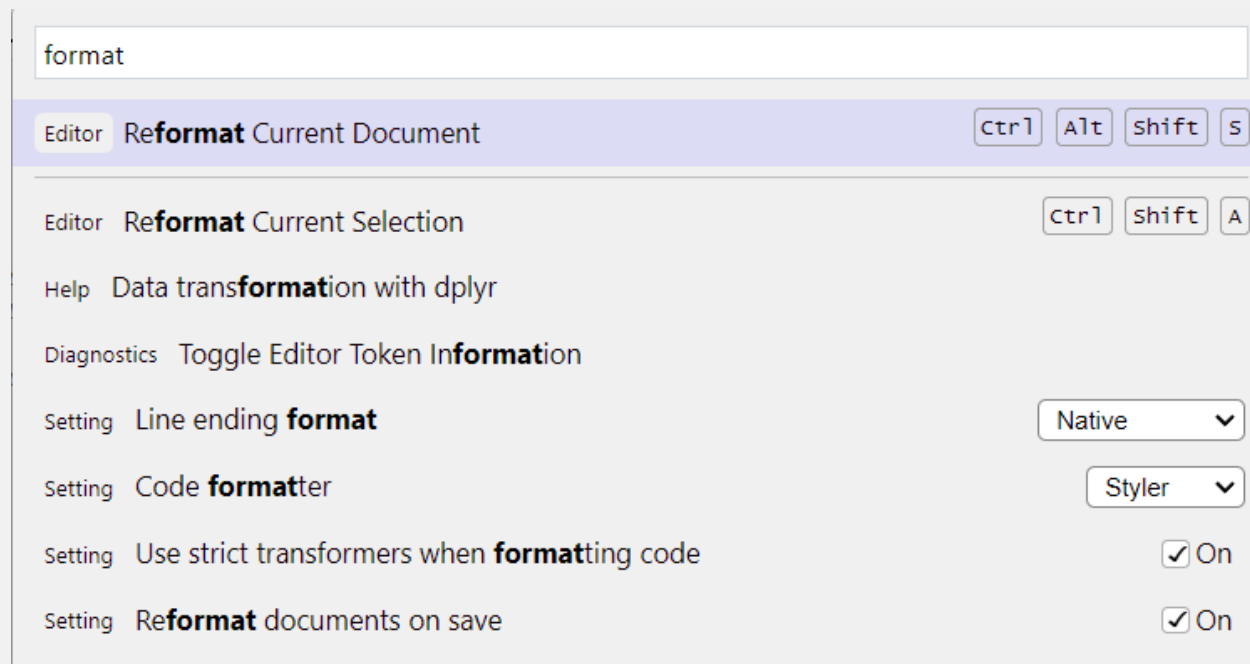
The console at the bottom shows the output of the lintr package, listing several linting errors for the script:

```
lintr
analysis/01_prepare_data.R
$ Line 6 [object_name_linter] Variable and function name style should be snake_case or symbols.
$ Line 10 [object_name_linter] Variable and function name style should be snake_case or symbols.
$ Line 10 [infix_spaces_linter] Put spaces around all infix operators.
$ Line 10 [infix_spaces_linter] Put spaces around all infix operators.
$ Line 10 [infix_spaces_linter] Put spaces around all infix operators.
$ Line 10 [line_length_linter] Lines should not be more than 80 characters.
$ Line 10 [spaces_inside_linter] Do not place spaces before parentheses.
```

Coding style helpers - Auto-formatting

IDEs offer auto-formatting tools.

- Auto-format your scripts on save and let the IDE do the job
- RStudio: Open command palette (**Tools** -> **Show command palette**), search for “format” -> “Reformat documents on save”



Modularize long scripts

One huge script is **hard to maintain**

- Break down long scripts into logical units
- Write scripts that do one thing, e.g.
 - `01_prepare-data.R`: Read raw data and prepare it for analysis
 - `02_run-models.R`: Run statistical analysis
 - `03_make-figures.R`: Create manuscript figures
- Call these scripts sequentially
- Use `source()` to source R scripts in other scripts

```
# source data preparation in other scripts  
source("01_prepare-data.R")
```

Modularize long scripts

Write a main **workflow script** that calls scripts in the right order.

- Often called `make.R`, `run.R` or `main.R`

```
# Prepare the data
source("01_prepare-data.R")
# Run all statistical models
source("02_run-models.R")
# Create manuscript figures
source("03_make-figures.R")
```

- Gives a nice overview of your workflow
- Very user-friendly: No need to run scripts separately and in the right order

Don't repeat yourself (DRY)

Example: Same data preparation code for multiple data sets

```
# Read the data
my_data1 <- readr::read_csv("data/data1.csv")
my_data2 <- readr::read_csv("data/data2.csv")
# Clean and summarize data
my_data1 <- my_data1 |>
  summarize(
    height = mean(height),
    biomass = mean(biomass),
    .by = c(country, species)
  )
my_data2 <- my_data2 |>
  summarize(
    height = mean(height),
    biomass = mean(biomass),
    .by = c(country, species)
  )
```

What's the **problem**?

Don't repeat yourself (DRY)

Solution: If you notice that you copy-paste code - write a **function**

Function in `R/prepare_data.R`:

```
prepare_data <- function(file_path) {  
  # Read the data  
  my_data <- read_csv(filepath)  
  # Clean and summarize data  
  my_data <- my_data |>  
    summarize(  
      height = mean(height),  
      biomass = mean(biomass),  
      .by = c(country, species)  
    )  
  # Return the cleaned data from the function  
  return(my_data)  
}
```

Source and use the `prepare_data` function in all scripts where it is needed

```
# Source the function  
source("R/prepare_data.R")  
  
# Use the function to read and prepare the data  
my_data1 <- prepare_data("data/data1.csv")  
my_data2 <- prepare_data("data/data2.csv")
```



Manage dependencies

Manage dependencies manually

Add the output of `devtools::session_info()` to your README

```
devtools::session_info()
```

```
— Session info —
setting  value
version  R version 4.4.2 (2024-10-31 ucrt)
os       Windows 11 x64 (build 26100)
system   x86_64, mingw32
ui       RTerm
language (EN)
collate  English_Germany.utf8
ctype    English_Germany.utf8
tz       Europe/Berlin
date     2025-06-19
pandoc   3.6.3 @ C:\\Users\\Selina\\AppData\\Local\\Programs\\Quarto\\bin\\tools/ (via rmarkdown)

— Packages —
package    * version date (UTC) lib source
cachem     1.1.0   2024-05-16 [1] CRAN (R 4.4.2)
cli        3.6.3   2024-06-21 [1] CRAN (R 4.4.2)
devtools   2.4.5   2022-10-11 [1] CRAN (R 4.4.2)
```


Manage dependencies with {renv}

Idea: Have a **project-local environment** with all packages needed by the project

- Keep log of the packages and versions you use
- Restore the local project library on other machines



Why this is useful?

- Code will still work even if packages upgrade
- Collaborators can recreate your local project library with one function
- Explicit dependency file states all dependencies

Check out the [renv website](https://rstudio.github.io/renv/) for more information

Manage dependencies with {renv}

```
# Get started  
install.packages("renv")
```

Very simple to use and integrate into your project workflow:

```
# Step 1: initialize a project level R library  
renv::init()  
# Step 2: save the current status of your library to a lock file  
renv::snapshot()  
# Step 3: restore state of your project from renv.lock  
renv::restore()
```

- Others only need to install the `renv` package, then they can also call `renv::restore()`

Conclusion

Take aways

- So many tips and guidelines 🌀
- Prioritize progress over perfection
 - Implement simple things first (e.g. format on save)
 - Other guidelines will become habits over time
- Plan ahead

Clean projects and workflows ...

... help you to write robust and reproducible code.



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Outlook

Of course there is much that I left out:

- Proper documentation
- Version control with Git
- Containerization (e.g. Docker) for full reproducibility
- Using R packages to build a research compendium
- ...

But this is for another time

Next lecture

Topic t.b.a.

 19th June  4-5 p.m.  Webex

- For topic suggestions and/or feedback [send me an email](#)
- [Subscribe to the mailing list](#)

Thank you for your attention
:)

Questions?

References

- [What they forgot to teach you about R book](#) by Jenny Bryan and Jim Hester
- [Blogpost](#) by Jenny Bryan on good project-oriented workflows
- [R best practice blogpost](#) by Krista L. DeStasio
- [Book about coding style for R: The tidyverse style guide](#)
- [The Turing way book](#) General concepts and things to think about regarding reproducible research
- [renv package website](#)
- [lintr package website](#)